



M1.1 Unit Overview

Patterns and Sequences

Students use tables, graphs, and expressions to make predictions about **sequences**. They determine if sequences are **arithmetic**, **geometric**, or neither. They write **recursive** and **explicit definitions** for sequences with **constant differences** and **constant ratios** and interpret parts of the definitions in context.

📧 Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Comparing Models, Modeling with Functions, NS: Parallels Between Numbers and Functions

Explore: Patterns Found in Nature

(Optional) Launch the unit with a non-routine task to investigate the question, "What mathematical patterns can be seen in nature?"

Sub-Unit 1: Sequences (Lessons 1–4)

- Make predictions about patterns and sequences.
- Write recursive definitions of sequences that have a constant difference or a constant ratio.
- Determine if sequences have a constant difference, constant ratio, or neither.
- **Language Goal:** Describe features of sequences.

Sub-Unit 2: Variables and Relationships (Lessons 5–7 + Practice Day)

- Write and interpret explicit definitions for arithmetic and geometric sequences.
- **Language Goal:** Interpret the meaning of expressions in context.

Contextual Vocabulary

branches, pine cone, tiles, balloon, council, pollution, proposal

Standards

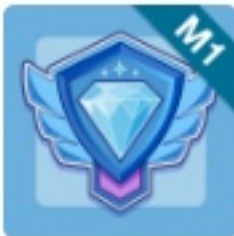
Addressing A-SSE.1 A-SSE.1.a F-BF.1.a F-LE.1.b F-LE.3 F-BF.2 F-LE.2 N-Q.3 SMP.1 SMP.2 SMP.3 SMP.4 SMP.5 SMP.7 SMP.8

Vocabulary

- arithmetic sequence
- constant difference
- constant ratio
- explicit definition
- geometric sequence
- model
- recursive definition
- sequence

Review Vocabulary

- expression
- linear relationship
- non-linear



Meet & Greet

Introduction to Math 1 and Each Other

This is intended to help you get to know your students and for them to get to know each other.

Note: This lesson is optional.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Today's Goals

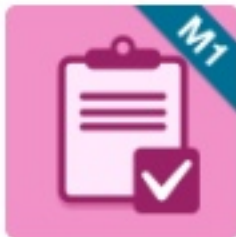
- Make connections with classmates and think about math collaboratively.
- Describe connections between the structure of a visual pattern and expressions that model it. (*Speaking and Listening*)

Standards

Addressing SMP.1 SMP.7

Instructional Routines

- Collect and Display (MLR2)
- Number Talk
- Think-Pair-Share



M1.1 Pre-Unit Check



Learn more about your students' understanding of foundational concepts and skills that will support them in the upcoming unit. This pre-unit check is intended for students to complete without a calculator.

OPTIONAL This is optional and can be given in its entirety before the unit begins or spread out throughout the unit.

📄 Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Number and Shape Patterns (Grade 4), Generalizing With Multiple Representations, Relationships Between Variables (Grade 6), CC3: Nets and Surface Area (Grade 6), Big and Small Numbers (Grade 8)

Standards

Assessing 4.OA.5 6.EE.1 6.EE.4 6.EE.6 6.EE.9 8.EE.1 SMP.2 SMP.3 SMP.7 SMP.8



Explore: Patterns Found in Nature

What mathematical patterns can be seen in nature?

OPTIONAL

This lesson is optional because this content extends beyond the scope of the standards.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Big Idea: CC2: Comparing Models

Today's Goal

- Language Goal:** Describe patterns seen in different representations of the Fibonacci sequence. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with **Building Math Habits of Mind** part of this Explore to support math language acquisition for your students.

Additional Vocabulary

Fibonacci sequence

Additional Materials

coloring tools

Standards

Building On 6.EE.9 Addressing SMP.6 SMP.7 Building Toward F-BF.1.a

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.5

Instructional Routines

- Notice and Wonder
- Discussion Supports (MLR8)



Visual Patterns

Lesson 1: Exploring Patterns

Let's explore visual patterns.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students receive Responsive Feedback as they create their own patterns using a fun and delightful interaction.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Comparing Models

Today's Goals

- Use tables and images to make predictions about a pattern.
- Language Goal:** Describe patterns in a table and in an image. (*Listening, Speaking, and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Additional Materials

scientific calculators (as needed)

Standards

Building On 6.EE.9 Addressing F-BF.1.a SMP.3 SMP.7 Building Toward F-LE.1.b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.5

Instructional Routines

- Number Talk
- Think-Pair-Share
- Discussion Supports (MLR8)



Sequence Carnival

Lesson 2: Introduction to Sequences

Let's explore sequences.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? This digital lesson uses Responsive Feedback to support students in visualizing the constant ratio and constant difference in several different sequences.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Comparing Models

Today's Goals

- Determine a missing term in a sequence.
- Language Goal:** Justify whether a sequence has a constant difference or a constant ratio. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 5.OA.3 Addressing F-BF.1.a SMP.3 SMP.8 Building Toward F-LE.1.b F-LE.1.c

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- constant difference
- constant ratio
- sequence

Instructional Routines

- Decide and Defend
- Notice and Wonder



Recursion Machine

Lesson 3: Recursive Definitions

Let's write recursive definitions of sequences to meet certain requirements.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Challenge Creator supports students in creating and sharing challenging math problems with classmates.

Instruction

Differentiation Beyond the Lesson

Big Ideas: NS: Parallels Between Numbers and Functions

Today's Goals

- Language Goal:** Define a sequence recursively to meet certain requirements. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 5.OA.3 Addressing F-BF.1.a F-LE.1.b F-BF.2 SMP.1 SMP.3 Building Toward F-LE.1.c F-LE.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- recursive definition

Instructional Routines

- Critique, Correct, Clarify (MLR3)



See the Sequence

Lesson 4: Arithmetic and Geometric Sequences

Let's compare sequences using tables and graphs.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students can focus on making generalizations about arithmetic and geometric sequences as the digital interactions create graphs and tables for them.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Comparing Models

Today's Goals

- Determine whether a sequence is *arithmetic*, *geometric*, or *neither*.
- Make connections between tables, graphs, and recursive definitions of sequences.
- Language Goal:** Compare *arithmetic* and *geometric* sequences using tables and graphs. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 8.F.5 Addressing F-LE.1.b F-LE.3 F-LE.2 SMP.3 SMP.5 SMP.8

Building Toward F-IF.4 F-IF.7 F-LE.1 F-LE.1.c F-LE.3 N-Q.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- arithmetic sequence
- geometric sequence

Review Vocabulary

- linear relationship
- non-linear

Instructional Routines

- Notice and Wonder
- Decide and Defend
- Discussion Supports (MLR8)



Paper Patterns

Lesson 5: Explicit Expressions

Let's represent situations in different ways.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Represent situations with tables, graphs, and expressions.
- Language Goal:** Interpret the meanings of parts of an expression in context. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Activity 3 Cards (one per pair, pre-cut); paper (optional); scissors (optional); Show What You Know Sheet

Standards

Building On 8.F.4 Addressing A-SSE.1 A-SSE.1a F-BF.1a F-BF.2 F-LE.2 SMP.1 SMP.7 SMP.8

Building Toward A-CED.1 A-CED.2 F-LE.1

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- explicit definition

Review Vocabulary

- expression

Instructional Routines

- Notice and Wonder
- Compare and Connect (MLR7)
- Think-Pair-Share



More Visual Patterns

Lesson 6: The n th Term

Let's write explicit expressions for arithmetic and geometric sequences.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Challenge Creator supports students in creating and sharing challenges with classmates.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Write an explicit expression for an arithmetic sequence or a geometric sequence.
- Understand that different explicit expressions can represent the same sequence.
- Language Goal:** Interpret the meanings of parts of an expression in context. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 7.EE.2 8.F.4 Addressing A-SSE.1 A-SSE.1.a F-BF.1.a F-BF.2 SMP.2 SMP.3 SMP.7 Building Toward A-CED.1 F-LE.1

Language Standards

ELD.PI.9-10.11 ELD.PI.9-10.3

Instructional Routines

- Compare and Connect (MLR7)
- Decide and Defend
- Critique, Correct, Clarify (MLR3)



Tree-Mendous Models

Lesson 7: Introduction to Modeling Situations

Let's model proposals mathematically and consider how they might affect a community.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Model situations using recursive definitions and explicit expressions to help solve real-world problems.
- Use math to make sense of issues in a community.
- **Language Goal:** Describe how to revise a model to incorporate new information. (*Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

CA EP&C Principle 1

Additional Materials

Show What You Know Sheet

Standards

Building On 8.F.4 Addressing A-SSE.1.a F-BF.1.a F-BF.2 F-LE.2 N-Q.3 SMP.1 SMP.2 SMP.3 SMP.4

Building Toward A-CED.1 F-LE.1

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6 ELD.PI.9-10.8

Vocabulary

- model

Instructional Routines

- Think-Pair-Share



M1.1 Practice Day

Students practice the concepts, skills, and strategies developed in this unit. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment.

📧 Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Comparing Models, Modeling with Functions, NS: Parallels Between Numbers and Functions

Additional Materials

Option 1: Level Up

Task cards: one set of cards per group

Option 2: Group Problems

Task cards: one set of cards per group

Standards

Addressing

A-SSE.1.a

F-BF.1.a

F-LE.1.b

F-BF.2

F-LE.2

SMP.2

SMP.7

SMP.8

Language Standards

ELD.PII.9-10.4



M1.1 Performance Task



Assign this summative assessment performance task at the end of the unit to evaluate students' proficiency with the concepts and skills addressed in the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Modeling with Functions, Comparing Models, NS: Parallels Between Numbers and Functions

Standards

Assessing

A-SSE.1.a

F-BF.1.a

F-BF.2

SMP.3

SMP.7



Unit Synthesis and Reflection



Here are six optional activities for students to engage in at the end of a unit to synthesize and/or reflect on their learning from the unit.

Notes:

- **Select one to two tasks** for students to complete before or after the End-Unit Assessment.
- Many tasks invite students to refer to the Student Goals and Glossary, which can be downloaded from the Unit Overview page at the beginning of each unit.

🗉 Let us know what you thought about this lesson by filling out [this survey](#).

Learning Goals

- Be purposeful, strategic, and goal-oriented about learning.
- Connect different ideas from the unit and across units.
- Revisit ideas from the unit in order to make revisions and reflect on their growth.